

# Nikhil Tammalla

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## SUMMARY

Final-year Computer Science student with hands-on experience in deep learning, computer vision, and applied AI. Proficient in Python, LLM pipelines, and medical image segmentation, with research published in IEEE and ICSSCC across 3 projects.

## EDUCATION

**Amrita Vishwa Vidyapeetham** — GPA: 7/10

*B.Tech in Computer Science Engineering*

**SR College** — GPA: 8.7/10

*Maths, Physics, Chemistry*

Bangalore, KA

May 2022 – June 2026

Warangal, TG

May 2020 – April 2022

## EXPERIENCE

**Data Intelligence & Analytics Intern**

*Anapty CodeEmy Technologies Private Limited, Coimbatore*

Feb 2026 – Apr 2026

- Developed and deployed ML models using scikit-learn and Python on 10,000+ records, achieving up to 85% accuracy; applied feature engineering and model evaluation techniques.
- Performed EDA and built data pipelines using Pandas and SQL, surfacing actionable insights to stakeholders through structured reporting.

## SKILLS

**Programming:** Python • JavaScript • SQL    **Databases:** MySQL • SQLite • Vector Databases (Qdrant)

**Machine Learning & AI:** scikit-learn • Supervised Learning • Classification • Regression • Feature Engineering • Model Evaluation

**Deep Learning & CV:** TensorFlow/Keras • PyTorch • CNNs • U-Net • Attention Mechanisms • Residual Networks • Medical Image Segmentation

**LLMs & Generative AI:** LLM Integration • RAG • Prompt Engineering • LangChain • Semantic Search • AI Chatbots

**Backend & Systems:** FastAPI • Node.js • REST APIs • WebSockets • Modular Backend Design    **Tools:** Git • Linux • VS Code

## PROJECTS

**Healthcare AI - Dental Assessment Chatbot** | *Node.js, Supabase, LLaMA 3.2B, Computer Vision* | GitHub

- Built an AI-driven dental assessment platform combining user inputs and intraoral images with an LLaMA 3.2B-powered chatbot, reducing preliminary diagnosis time by ~60%.
- Integrated appointment booking and patient history management via Supabase, supporting 3+ condition categories with sub-2s average response time.

**Brain Tumor MRI Segmentation & Classification** | *Python, Attention U-Net, CNNs, TensorFlow/Keras* | GitHub

- Built an end-to-end MRI analysis pipeline across 6,380 images (4 tumor classes) with 20+ hyperparameter tuning experiments to optimize loss functions.
- Achieved ~90% Dice score and ~95% classification accuracy using Attention ResUNet architecture with ensemble-based post-processing.

**RAG-Based LLM Chatbot for Document QA** | *Python, LLaMA 3.2, RAG, Qdrant, LangChain* | GitHub

- Built a RAG chatbot integrating LLaMA 3.2 with Qdrant vector embeddings, handling 1,000+ document chunks with ~88% answer relevance.
- Designed a semantic chunking and similarity search ingestion pipeline, reducing query response time by 40% over keyword-based retrieval.

## CERTIFICATIONS & PUBLICATIONS

**Certifications:** Data Analyst 101 — Simplilearn (2025)    —    Python Mega Course — Udemy (2023)    —    Data Visualization — Kaggle (2026)

### Publications:

- ResDenseAtt-UNet: An Attention-Enhanced DenseUNet Model for Dental Segmentation* — ICSSCC (2024)
- Real-Time Web Chat Application with Live Performance Metrics* — IEEE (2024)
- Healthcare AI - Dental Assessment Chatbot* — IEEE (2025)